

Jacob Marriott

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Profile

I am a fourth-year Software Engineering student at the University of Queensland who runs an independent software consulting practice, shipping full-stack web and AI systems to production for real clients. I work daily across the modern web stack (Next.js, React, TypeScript, FastAPI, and PostgreSQL) and build agentic AI systems using large language models, LangGraph, and retrieval-augmented generation, with a strong emphasis on testing, security, and reliable delivery. Proficient in Java, Python, C, and Haskell, I have experience across object-oriented, functional, and network programming. My background in professional kitchens has sharpened my ability to work under pressure, communicate clearly, and take ownership in fast-paced environments. I care about building software that is both technically robust and genuinely useful, and I bring adaptability, initiative, and strong collaboration skills to any engineering team.

Education

The University of Queensland — Bachelor of Engineering (Honours), Software Engineering
Brisbane, QLD | 2023 – Present

Wakatipu High School — Queenstown, New Zealand
2018 – 2022

Skills

- Programming Languages: Python, TypeScript/JavaScript, Java, C, Swift, Haskell, SQL
- AI & Agentic Systems: large language models, LangGraph, retrieval-augmented generation (RAG), pgvector, agent orchestration, prompt engineering
- Web & Full-Stack: Next.js, React, FastAPI, Node.js, Supabase/PostgreSQL, Vercel, Tailwind CSS
- Engineering Practices: Git, CI/CD, automated testing (Jest, Playwright, Vitest), test-driven development, security hardening (auth, CSRF, rate limiting, validation)
- Software Design: object-oriented, functional, and network programming
- Collaboration: project management, technical communication, and problem-solving

Selected Projects

- Brain Console – a unified, business-wide AI “brain”: a stateful agent with persistent memory and tool execution that centralises a company's context, knowledge, and workflows and automates tasks across its tools. My flagship AI-consulting product (Python, LangGraph, and pgvector).
- dAIly – a voice-first AI personal assistant built on LangGraph with a pgvector long-term memory layer and real-time speech (Deepgram and Cartesia); FastAPI backend, headed to production.

- Agentic OS – a persistent personal AI runtime: a Slack-driven orchestrator that coordinates specialist agents over my projects, state, and workflows (Python, FastAPI, and scheduling).

Volunteering

Member, Electrically Based Engineering Student Society – UQ Engineering Undergraduate Society (2024 – Present)

Member, Student Staff Liaison Group **-** School of Electrical Engineering and Computer Science (2025 - Present)

Student Representative, Chairperson – SSLG:
Software Engineering Studio: Design, Implement and Test (Semester 2, 2025)
Software Architecture (Semester 1, 2026)
Generative Artificial Intelligence (Semester 1, 2026)

Qualifications

- **SITXFSA005** – Use Hygienic Practices for Food Safety (2024–Present)
- **HLTAID009** – Provide Cardiopulmonary Resuscitation (2024–Present)
- **HLTAID011** – Provide First Aid (2024–2027)

Work Experience

Software Engineer (Independent / Contract) | Brisbane, QLD
Mar 2026 – Present

- Chef Jon Pizzeria & Bistro – built a full-stack hospitality operations dashboard (Next.js, React, TypeScript, and Supabase) with real-time P&L, sales, labour, and COGS tracking, PIN/cookie auth, Chart.js visualisations, 227 automated tests, and security hardening (CSRF, rate limiting, and request validation); deployed on Vercel with CI/CD, now integrating Xero via OAuth2.
- Innovative Travel – built serverless middleware (TypeScript, Vercel Edge Functions, and Upstash Redis) syncing a Shopify storefront into a travel-agency booking feed, with an admin pricing/FX dashboard and session-based auth.
- In A Pickle – delivered and now maintain a multi-page marketing site with a custom lightweight CMS (Node.js, Vercel Blob, Resend, and Cloudflare Turnstile) for menu, blog, and content updates.
- Sandbox Alpha (AI consulting) – lead engineer designing and rolling out a company-wide AI intelligence layer across a multi-business portfolio (see Selected Projects).
- Work extensively with agentic coding tools (Claude Code) to accelerate delivery while personally reviewing all generated code.

Chef de Partie – Chef Jon Pizzeria & Bistro | Brisbane, QLD
Aug 2025 – Present

- Oversee daily operations of the bistro section, ensuring precision and efficiency under pressure
- Train, supervise, and motivate kitchen staff while fostering teamwork and accountability
- Program and modify POS systems to maximise efficiency and communication between FOH and BOH

Chef de Partie – Boucher French Bistro | Brisbane, QLD
Dec 2025 – Mar 2026

Commis Chef – Ruby, My Dear | Brisbane, QLD
Jan 2025 – Aug 2025

- Supported menu development and recipe refinement through iterative experimentation
- Managed prep operations independently, maintaining consistent quality under time constraints

Commis Chef – Tanoshi Restaurant Group (Part-Time) | Queenstown, NZ
Nov 2021 – Dec 2025

- Implemented traditional Japanese cooking techniques, mastering precision and timing
- Maintained hygiene and safety compliance while training junior staff

Sound Engineer – Rock Formation Trust (Contract) | Queenstown, NZ
Nov 2020 – Dec 2022

References

Dr. Guowei Yang – Senior Lecturer, School of Electrical Engineering and Computer Science
guowei.yang@uq.edu.au

Matty Yates – Founder / Managing Director, Tanoshi Restaurant Group
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Casey Poland – Former Executive Chef, Happy Fat Group (Ruby, My Dear)
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